Melanoma Classification

Deep Learning Project Report

Machine Learning and Data Mining 2 (MLDM2) · FS26
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This report documents the development of a deep learning pipeline for automated melanoma classification using the SIIM-ISIC Melanoma Classification 2020 dataset. The task involves multimodal data (skin lesion images combined with clinical patient metadata) and a severe class imbalance of approximately 2% positive cases. We implement and compare a multimodal model against an image-only baseline, with a focus on architecture decisions, class imbalance handling, threshold analysis, and clinical interpretation of results.

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1. Problem Statement

Melanoma is the most dangerous form of skin cancer. Early and accurate detection is critical for patient outcomes, as survival rates drop significantly with late-stage diagnosis. The SIIM-ISIC 2020 competition provides a curated dataset of dermoscopic skin lesion images alongside structured patient metadata, making it an ideal benchmark for multimodal deep learning research.

The primary challenge is a severe class imbalance: approximately 98% of samples are benign (non-cancerous) and only ~2% are malignant (melanoma). Standard accuracy metrics are therefore misleading — a model predicting "benign" for every sample achieves 98% accuracy which makes it clinically useless. This motivates the use of AUC-ROC as the primary metric and careful threshold selection informed by clinical priorities.

In a medical context, False Negatives (missed melanomas) are far more harmful than False Positives (unnecessary follow-up). This asymmetry directly influences our choice of loss function, sampling strategy, and classification threshold.

2. Dataset

2.1 Overview

We use the SIIM-ISIC Melanoma Classification 2020 dataset, accessed via Kaggle. Due to the large size of the original dataset (116 GB), we use a pre-processed version with images resized to 224x224 pixels (~3.5 GB). The dataset contains 33'126 training samples with labels.

Property	Value
Total training samples	33'126
Positive cases (melanoma)	~584 (1.76%)
Negative cases (benign)	~32,542 (98.24%)
Image resolution	224 x 224 pixels (RGB)
Metadata features	Age, sex, anatomical site
Split (train / val / test)	70% / 15% / 15% — stratified

2.2 Train / Val / Test Split

To ensure a representative distribution of melanoma cases across all subsets, we applied stratified splitting. Since only ~1.76% of samples are positive, a purely random split could result in validation or test sets with very few — or even zero — melanoma cases, making evaluation unreliable. Our final split is 70% training (23,188 samples), 15% validation (4,969), and 15% test (4,969), each maintaining the original class ratio."

All preprocessing parameters (age normalization bounds) are fitted exclusively on the training set and applied to validation and test sets to prevent data leakage.

3. Data Preprocessing & Feature Engineering

3.1 Image Preprocessing

All images are resized to 224x224 pixels to match the input requirements of the EfficientNet-B0 backbone. Pixel values are normalized using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]), since the backbone was pretrained on ImageNet.

During training, the following augmentations are applied to improve generalization: random horizontal and vertical flips, random rotation (±15°), and color jitter (brightness, contrast, saturation ±0.2). No augmentation is applied during validation or testing to ensure fair evaluation.

3.2 Metadata Feature Engineering

Three clinical features are extracted and encoded as numerical vectors for the metadata branch of the model:

Feature	Encoding	Dim
Age (age_approx)	Min-max normalization [0,1] — fitted on training set only	1
Sex	One-hot encoding: male=[1,0], female=[0,1]	2
Anatomical site	One-hot: torso, lower/upper extremity, head/neck, palms/soles, oral/genital	6
Total metadata vector		9

Missing values are filled with 0 before encoding. Age normalization bounds are computed on the training set only (min=0, max=90), then reused for validation and test preprocessing.

4. Model Architecture

4.1 Multimodal Model

Our main model processes image and metadata in parallel branches before fusing them into a single classification head. This design prevents the 9-dimensional metadata vector from being overwhelmed by the 1,280-dimensional image feature vector.

Branch	Component	Output Dim
Image	EfficientNet-B0 (pretrained ImageNet)	1,280
Image	Projection: Linear(1280→128) + BatchNorm + ReLU + Dropout(0.3)	128
Metadata	MLP: Linear(9→64) + ReLU + Linear(64→32) + ReLU	32
Fusion	Concatenation of image + metadata features	160
Classifier	Linear(160→64) + ReLU + Dropout(0.3) + Linear(64→1)	1 (logit)
Output	Sigmoid → probability [0, 1]	—

Total trainable parameters: ~4.2M. The EfficientNet-B0 backbone uses pretrained ImageNet weights and is fine-tuned end-to-end. Transfer learning significantly accelerates convergence and improves generalization on the relatively small positive class.

4.2 Baseline Model (Image Only)

For comparison, we train an image-only baseline using the same EfficientNet-B0 backbone with a direct classifier head (no metadata branch). This allows us to isolate the contribution of clinical metadata to model performance.

4.3 Training Configuration

Hyperparameter	Value

Loss function	BCEWithLogitsLoss with pos_weight=50
Optimizer	Adam, lr=1e-4
Batch size	32
Max epochs	30 stopped at 7 (patience=5)
Sampler	WeightedRandomSampler (oversamples positives)
Dropout	0.3
Hardware	NVIDIA T4 GPU (LightningAI)
Framework	PyTorch Lightning

The training history confirms stable convergence: train loss decreases consistently across all steps, while validation AUC-ROC plateaus around 0.90 after approximately 6 epochs. The increasing validation loss alongside stable AUC suggests the model begins to overfit in later epochs, which justifies the early stopping strategy used during training

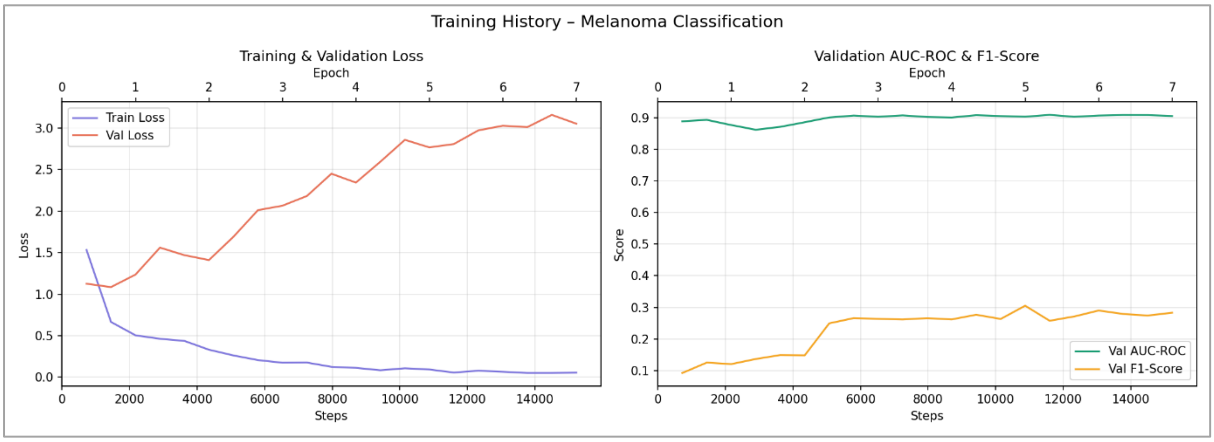

Figure 1: Training and validation loss (left) and validation AUC-ROC & F1-Score (right) over epochs for the multimodal model.

5. Handling Class Imbalance

With only ~1.76% positive cases, class imbalance is the central challenge of this task. We address it with two complementary strategies:

5.1 Weighted Loss Function

BCEWithLogitsLoss with pos_weight=50 penalizes missed melanomas (False Negatives) approximately 50 times more heavily than False Positives. This ratio is motivated by the inverse class frequency (98/2 ≈ 49) and by the clinical asymmetry: a missed melanoma has far more severe consequences than an unnecessary follow-up appointment.

5.2 WeightedRandomSampler

During training, melanoma cases are oversampled using PyTorch's WeightedRandomSampler. Each sample is assigned a weight inversely proportional to its class frequency, ensuring the model sees approximately equal numbers of positive and negative examples per batch.

5.3 pos_weight Hyperparameter Search

We systematically evaluate three values of pos_weight to understand the Sensitivity–Specificity trade-off:

pos_weight	AUC-ROC	Sensitivity	Specificity	TP	FN	FP
25	0.9124	0.402	0.977	35	52	112
50 (selected)	0.9126	0.632	0.932	55	32	331
75	0.9182	0.598	0.939	52	35	297

pos_weight=50 is selected as the best trade-off: it achieves the highest Sensitivity (0.632) — the fewest missed melanomas — while maintaining a competitive AUC of 0.9126.

6. Evaluation & Results

6.1 Multimodal vs. Baseline (Threshold = 0.50)

Both models are evaluated on the held-out test set. AUC-ROC is used as the primary metric because it is threshold-independent and robust to class imbalance.

Metric	Multimodal	Baseline (image only)
AUC-ROC	0.9126	0.9090
F1-Score	0.2326	0.2374
Sensitivity (Recall)	0.6322	0.6207
Specificity	0.9322	0.9357
True Positives (TP)	55	54
False Negatives (FN)	32	33
False Positives (FP)	331	314

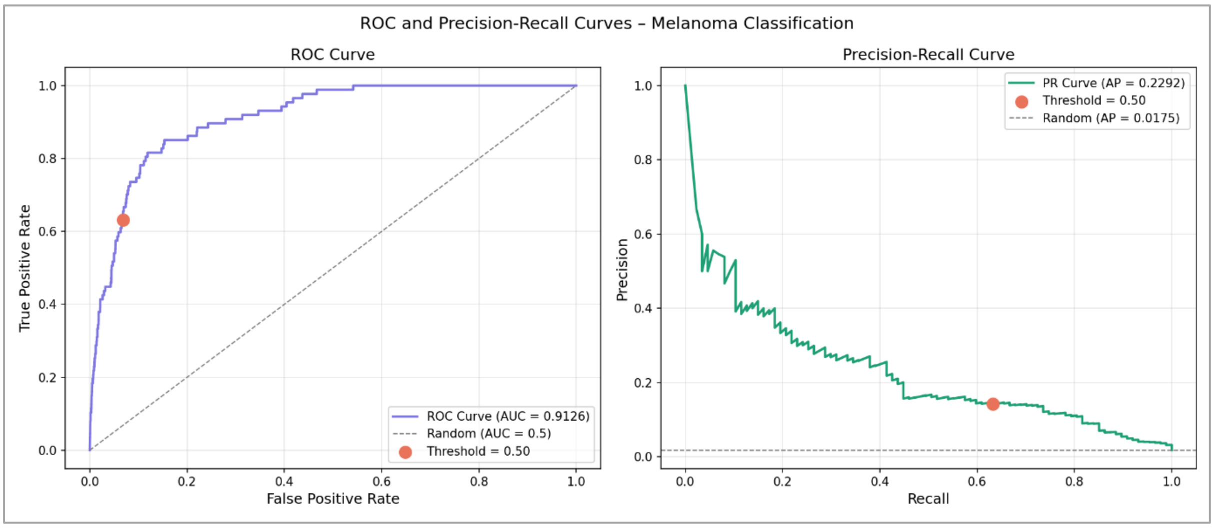

Figure 2: ROC Curve (left) and Precision-Recall Curve (right) of the multimodal model on the test set

The multimodal model achieves higher AUC-ROC and Sensitivity, meaning it identifies more true melanomas. The baseline achieves marginally better Specificity and F1-Score. Overall, image features dominate the prediction and metadata provides a small but consistent improvement. This is consistent with findings in the literature on this dataset.

6.2 Threshold Analysis

The default threshold of 0.50 is not necessarily optimal for medical screening. By lowering the threshold, we accept more False Positives in exchange for fewer missed melanomas. The table below shows this trade-off for the multimodal model:

Threshold	Sensitivity	Specificity	F1-Score	TP	FN	FP
0.05	0.782	0.894	0.202	68	19	517
0.10	0.736	0.907	0.212	64	23	452
0.20	0.736	0.917	0.229	64	23	407
0.30	0.701	0.922	0.230	61	26	382
0.50 (default)	0.632	0.932	0.233	55	32	331

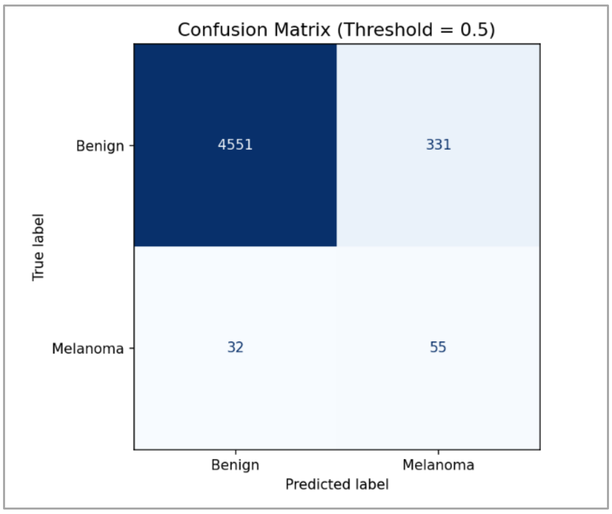

Figure 3: Confusion matrices at default threshold=0.5

At threshold 0.05, Sensitivity reaches 0.782 — the model detects 68 out of 87 melanomas. In a real-world screening scenario, thresholds of 0.05–0.10 would be preferred: missing a melanoma has life-threatening consequences, while a false alarm results only in a follow-up biopsy.

6.3 Error Analysis

False Positives tend to be atypical benign lesions with irregular pigmentation or borders, which visually resemble melanoma. False Negatives are the most clinically critical errors, often appearing as early-stage melanomas with subtle features. Lowering the classification threshold directly reduces this error category at the cost of more false alarms.

Figure 4 reveals systematic error patterns across body site, age, sex, and model confidence. False Negatives (missed melanomas) occur most frequently on the torso, upper extremity, and lower extremity body sites, which also contains the majority of melanoma cases in the dataset, suggesting these errors reflect class difficulty rather than a systematic site specific bias. The age distribution shows False Negatives are spread across all age groups, with a slight concentration in the 50–70 range. The confidence histogram is particularly revealing that False Negatives cluster near p=0.00, meaning the model was highly confident in its incorrect benign prediction. These are the most dangerous errors clinically.

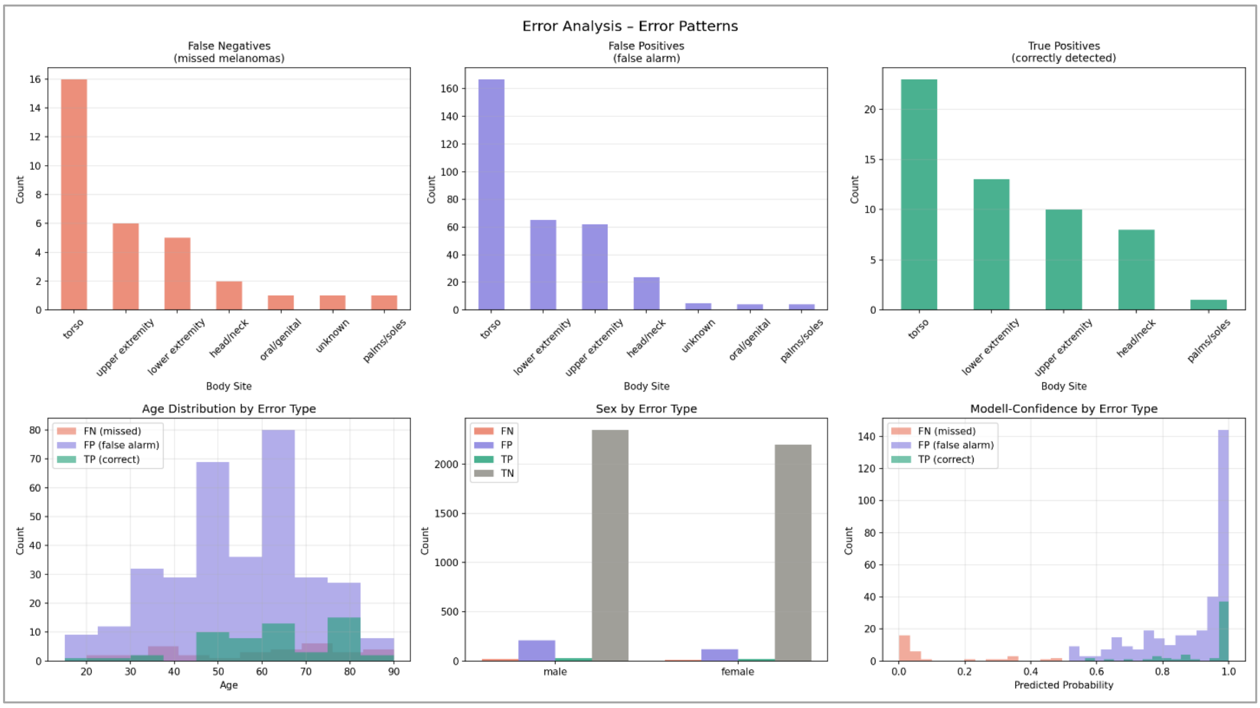

Figure 4: Error pattern analysis by body site, age, sex, and model confidence

Figures 5 show the most confidently misclassified images. False Negatives (left) often present as subtle, early-stage lesions with uniform coloration and regular borders. False Positives (right) tend to be atypical benign lesions with irregular pigmentation, asymmetric borders, or dark pigment clusters that visually mimic melanoma. These patterns highlight the inherent difficulty of dermoscopic classification even for trained models, and motivate the use of a lower classification threshold in clinical screening to reduce the number of missed melanomas

Figure 5: FN examples (left) ; FP examples (right)

7. Discussion

7.1 Multimodal vs. Baseline

The multimodal model consistently outperforms the baseline on AUC-ROC and Sensitivity, though the gains are modest. Patient metadata (age, sex, body site) provides complementary signal to image features, but cannot overcome the dominance of visual information in dermoscopy. Finer clinical variables (e.g. lesion history, patient risk factors) might provide stronger multimodal gains.

7.2 Class Imbalance

The combination of pos_weight and WeightedRandomSampler proved effective: the model avoids collapsing to a trivial all-negative prediction while maintaining competitive AUC. The optimal pos_weight of 50 aligns with the inverse class frequency, suggesting that a data-driven initialization of this hyperparameter is a reasonable starting point.

7.3 Limitations

The model does not achieve the target Sensitivity of ≥0.80 at threshold 0.50; this threshold must be lowered to ~0.05 to approach clinically acceptable recall, at significant cost to Specificity. F1-scores in the 0.20–0.24 range reflect the inherent difficulty of this imbalanced task and are consistent with other published solutions on this dataset.

8. Conclusion

We developed and evaluated a multimodal deep learning pipeline for melanoma classification combining EfficientNet-B0 image features with clinical metadata. Our model achieves an AUC-ROC of 0.9126, outperforming the image-only baseline (0.9090) and demonstrating that clinical metadata provides consistent marginal improvements.

Key contributions: (1) a dual-branch architecture preventing metadata from being overwhelmed by high-dimensional image features; (2) a combined imbalance strategy using pos_weight and WeightedRandomSampler; (3) a systematic hyperparameter search over pos_weight values; (4) a threshold analysis demonstrating the clinical relevance of threshold selection for medical screening tasks.

For clinical deployment, we recommend a threshold of 0.05–0.10 to maximize Sensitivity and minimize missed melanomas, accepting the associated increase in False Positives as a clinically justified trade-off.

References

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[5] Pre-processed 224x224 images: https://www.kaggle.com/datasets/arroqc/siic-isic-224x224-images

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